S. VIKAS

Systems Software & Edge AI Engineer

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Professional Summary

Systems Software and Edge AI Engineer (B.Tech CSE, 2027) with a deep technical focus on **bare-metal C++17, Linux kernel memory management, lock-free concurrency, and deterministic GPU/edge runtimes**. Architected **EdgeModelScheduler v4.0**, an open-source C++17 deterministic co-scheduler for constrained edge SoCs with zero-copy CUDA IPC (12.23 GB/s), sub-15ns lock-free SPSC rings, and an ISO 26262 ASIL-D flight recorder. Hands-on R&D experience at **Ashok Leyland (Guided by DGM)** developing embedded C++ control systems on physical robots handling high-frequency LiDAR and encoder sensor streams. Winner of the national Agentic AI Hackathon 2025, with three patents filed and two research papers under review.

Education

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai 2023 – 2027
B.Tech – Computer Science and Engineering ***CGPA: 8.67 / 10***

St. Mary's Matriculation Boys Higher Secondary School – Class XII, Tamil Nadu State Board 80.33% (2023)

St. Mary's Matriculation Boys Higher Secondary School – Class X, Tamil Nadu State Board 100.00% (2021)

Technical Skills

- **Programming Languages:** C++, C++17, C, Java (Java 21), Python, SQL, JavaScript (ES6+), TypeScript, Bash/POSIX Shell, PHP
- **Low-Level & Systems Engineering:** Bare-Metal C++, POSIX Threads, Lock-Free SPSC/MPSC Queues (`alignas(64)`), Cache-Line Optimization, Linux Shared Memory (`/dev/shm`, `mmap`), Zero-Copy CUDA IPC (12.23 GB/s), Atomics, Memory Barriers
- **AI & Acceleration Runtime:** CUDA, TensorRT, ONNX Runtime, Deep Learning Accelerators (DLA), PyTorch, OpenCV, YOLOv8, YOLOv12 (Ultralytics), Quantization (INT8/FP16, RKNN), Thermal DVFS Governors, Ollama, Azure AI Foundry, AI Agents
- **Embedded & Hardware Systems:** Teensy 4.1, Rock 5 ITX, Raspberry Pi, GPIO (`gpiozero`), High-Frequency LiDAR Point Clouds, Rotary Encoders, Hardware Relays, Closed-Loop Motor Actuation
- **Backend & Enterprise Stack:** Java 21, Spring Boot, Spring Security, Spring Data, REST API Design, STOMP over WebSocket, JWT Authentication, BCrypt, Node.js, Express.js, Flask
- **Databases & DevOps:** MySQL, MSSQL, MongoDB (Atlas), Firebase; Linux (Ubuntu/Debian), Git, GitHub Actions (CI/CD), Docker, CMake, GTest, Postman
- **Core Computer Science:** Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, Software Engineering, System Design, SDLC

Professional Experience

Project Intern Mar 2024 – Dec 2024
Ashok Leyland (Guided by DGM) *Chennai, India*

- Owned the **end-to-end lifecycle** of an autonomous agricultural robot, from conceptual design and system architecture to a working hardware prototype under DGM mentorship, directly contributing to a **patent filing** and research publications.
- Engineered embedded **C++** control logic on a Teensy 4.1 micro-controller handling high-frequency **LiDAR** and optical encoder sensor streams for deterministic real-time actuation and closed-loop obstacle avoidance.
- Quantized and deployed a custom **YOLOv8** model via **RKNN** on a Rock 5 ITX, achieving sub-40ms edge inference and real-time weed classification in field environments.

AI Engineer Intern Aug 2025 – Oct 2025
Siter Academy, Norway (Remote) *Oslo, Norway*

- Built **MedPal AI**, a transformer-based medical assistant combining Generative AI, Computer Vision, NLP, and Document Intelligence, as part of a team project under an industry research mentor.
- Engineered an OCR pipeline (Tesseract, OpenCV) for pill recognition and integrated a locally-hosted LLM (Ollama) into Flask to preserve healthcare data privacy.
- Completed applied labs on Azure AI Foundry: built multi-agent solutions with Semantic Kernel SDK, text classification, and speech-to-text pipelines; applied Responsible AI safeguards against adversarial prompts.

Web Development Intern Aug 2025 – Present
Office of Dean Academics, Vel Tech University *Chennai, India*

- Architected a centralized **Campus Recruitment Portal** (miniapps.veltech.edu.in/c2c) on a **dual-database design (MSSQL and MySQL)**, building a secure synchronization layer to reconcile academic records across both systems without data loss.
- Developed an automated eligibility engine and a custom Applicant Tracking System (ATS) using rule-based SQL queries and PDF resume parsing, eliminating manual shortlisting for 1000+ student records.
- Engineered a role-based Research Studies Management System (miniapps.veltech.edu.in/research_studies) with a dynamic timeline engine computing stage deadlines, including fractional-month logic, for PhD scholar progress tracking.

Key Projects

EdgeModelScheduler v4.0 – Deterministic Edge AI Co-Scheduler github.com/VIKASSAMVEL/edge-model-scheduler
Tech Stack: C++17, CUDA IPC, Linux /dev/shm, Lock-Free Rings, CMake vikassamvel.github.io/edge-model-scheduler

- Architected a pure C++17 deterministic edge AI co-scheduler for constrained edge SoCs (NVIDIA Jetson / automotive platforms), operating with an ultra-lean 3.125 MB VmRSS footprint (**∼396x leaner** than Triton Inference Server's 1,240 MB idle footprint).
- Engineered a CUDA IPC Zero-Copy TypeAdapter transferring raw LiDAR point clouds and camera frames across process boundaries via 64-byte virtual handles over `/dev/shm` at **12.23 GB/s throughput (<15 μs mapping latency)**, eliminating ROS 2 DDS serialization overhead.
- Implemented sub-15ns lock-free Single-Producer Single-Consumer (SPSC) circular rings with head and tail atomics isolated on separate 64-byte cache lines (`alignas(64)`), eliminating CPU lock contention and cache-line bouncing.
- Built slack-aware pre-execution shedding that continuously evaluates remaining frame deadlines before GPU submission, bounding P99 worst-case tail latency to **29.02 ms** (vs. 73 ms+ in unmanaged FIFO) with **0.00% deadline miss rate** under simulated 45 ms network jitter stalls.
- Integrated an ISO 26262 ASIL-D telemetry flight recorder (60,000 frames @ 1 kHz ring buffer) auto-dumping rolling SHA-256 cryptographically verified `.emsbag` binaries on safety trips, paired with an 82°C thermal DVFS hysteresis governor.

Autonomous Agricultural Robot – Edge AI & Real-Time Control (Ashok Leyland) **Patent Filed (2024)**
Tech Stack: C++, Python, YOLOv8, RKNN, Rock 5 ITX, Teensy 4.1, LiDAR, Optical Encoders

- Deployed an INT8 quantized YOLOv8 model on a Rock 5 ITX for weed classification, interfaced with C++ embedded control logic on a Teensy 4.1 micro-controller handling high-frequency LiDAR and encoder sensor streams for real-time actuation. **Patent filed.**
- Engineered deterministic pulse-width modulation motor actuation routines, achieving sub-millisecond control loop response times.

Construction Worker Safety System – Edge AI & Vision (Ashok Leyland) **Patent Filed (2025)**
Tech Stack: Python, YOLOv12 (Ultralytics), OpenCV, PyQt5, Raspberry Pi, GPIO, EdgeTAM

- Built a real-time computer vision system detecting six safety-critical classes (workers, harnesses, helmets, ladders, scaffolding, MEWPs) to monitor work-at-height compliance on construction sites.
- Implemented spatial safety logic: **IOU-based harness-compliance checks** and point-in-polygon zone-violation detection with hysteresis debouncing; wired GPIO relay on Raspberry Pi to trigger sirens or cut equipment power upon hazardous violations. **Patent filed.**

E2EE Chat System – Secure Real-Time Messaging Platform github.com/VIKASSAMVEL/E2EE
Tech Stack: Java 21, Spring Boot, Spring Security, Spring Data MongoDB, STOMP/WebSocket, React, Docker

- Architected a layered Spring Boot backend with stateless JWT authentication, custom `AuthTokenFilter`, and method-level security; engineered real-time delivery over STOMP messaging with a custom `ChannelInterceptor` for principal binding.
- Implemented **hybrid envelope encryption** on the client (RSA-OAEP 2048-bit key wrapping with ephemeral AES-GCM 256-bit session key per message) via WebCrypto API; server persists only ciphertext, guaranteeing zero-knowledge security. Containerized with Docker.

Krishi Jarvis – Multilingual Farm Advisory & Plant Disease Detection github.com/VIKASSAMVEL/farmer-agent
Tech Stack: Flutter, Flask, YOLOv8 (ONNX Runtime), Local Ollama LLM, Google Translate API

- Trained and deployed a YOLOv8 model (exported to ONNX) to classify 29 leaf-disease classes from camera images; built local Ollama LLM RAG layer grounded in verified agronomy data, offering multi-lingual voice/text interaction in Hindi, Tamil, and English.

Achievements

- **Winner** – Agentic AI Hackathon 2025, SIMATS (world's first prompt-exclusive hackathon).
- **National Rank 37 of 150** – TechXcelerate Hackathon, BITS Goa.
- **Top 5%** – ODOO x MSU FootPrints'25 Hackathon.
- **First Prize** – Technical Paper Presentation Competition, ISTE Tamil Nadu (PSR College).
- **Qualifier** – Smart India Hackathon (SIH) Internal Institutional Rounds, 2024 and 2025.

Patents and Publications

- **Patent Filed:** Automatic Railway Gate System (Indian Patent Office, 2023).
- **Patent Filed:** Autonomous Agricultural Robot, Ashok Leyland (Indian Patent Office, 2024).
- **Patent Filed:** Real-Time Construction Worker Fall Prevention and Safety Compliance Monitoring System, Ashok Leyland (Indian Patent Office, 2025).
- **Research (Under Review):** Real-Time Autonomous Navigation for Agricultural Robots.
- **Research (Under Review):** Indian Sign Language Recognition using Deep Learning.